

Program 8

Step 1:- Create [server.js](#), nodejs-configmap.yaml and nodejs-pod.yaml

```
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ gedit server.js
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ gedit nodejs-configmap.yaml
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ gedit nodejs-pod.yaml
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ la
nodejs-configmap.yaml  nodejs-pod.yaml  server.js
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ cat server.js
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running on Kubernetes!");
});

server.listen(port, () => {
  console.log('Server running at port ${port}');
});
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ cat nodejs-
nodejs-configmap.yaml  nodejs-pod.yaml
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ cat nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;
    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });
    server.listen(port, () => {
      console.log('Server running at port ${port}');
    });
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ cat nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
  - name: nodejs
    image: node:18
    command: ["node", "/usr/src/app/server.js"]
    volumeMounts:
    - name: app-code
      mountPath: /usr/src/app
    ports:
      - containerPort: 3000
  volumes:
  - name: app-code
    configMap:
      name: nodejs-app-config
```

Step 2:- Apply the yaml files

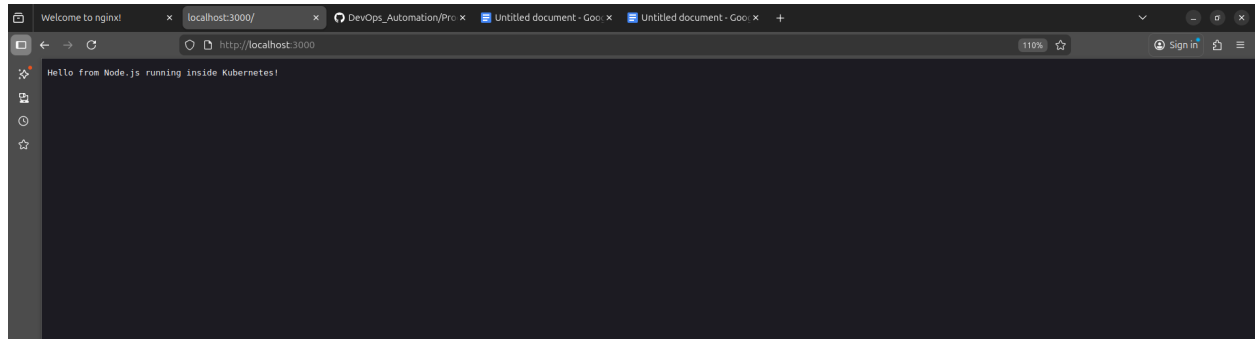
```
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
```

Step 3:- Check pod status

```
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-bprbr   1/1     Running   0           11m
nginx-deployment-6f9664446b-bvhgj   1/1     Running   0           11m
program-8                            1/1     Running   0           38s
```

Step 4:- Access Nodejs app

```
1rv24nc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```



Step 5:- Delete all pods and deployments once executed

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-bprbr" deleted from default namespace
pod "nginx-deployment-6f9664446b-bvhgj" deleted from default namespace
pod "program-8" deleted from default namespace

1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
service "nginx-deployment" deleted from default namespace
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted from default namespace
```